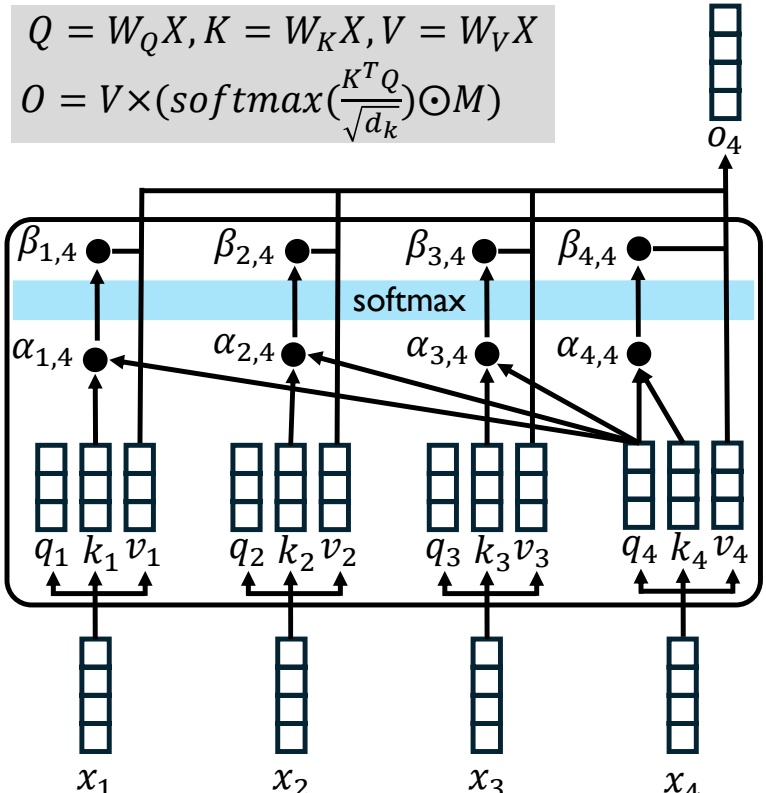
$$Q = W_Q X, K = W_K X, V = W_V X$$

$$O = V \times (softmax(\frac{K^T Q}{\sqrt{d_k}}) \odot M)$$



$$o_t = \sum_{j=1}^{t} \beta_{j,t} v_j$$

$$\beta_{j,t} = \frac{\exp(\alpha_{j,t})}{\sum_{j=1}^{t} \exp(\alpha_{j,t})}$$

$$\alpha_{j,t} = \frac{k_j^T q_t}{\sqrt{d_k}}$$

$$q_t = W_Q x_t \ (W_Q \in R^{d_k \times d})$$

$$k_t = W_K x_t \ (W_K \in R^{d_k \times d})$$

$$v_t = W_V x_t \ (W_V \in R^{d_v \times d})$$